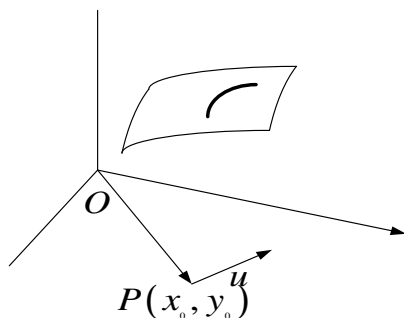


§14.6 Directional Derivatives and Gradient Vector

Homework : 9,17,19,23,31,35,43,47,53,59

(1) 方向導數(沿著 $u = \langle a, b \rangle$, $a^2 + b^2 = 1$, 方向的微分.)



定義 : $u = \langle a, b \rangle$, $a^2 + b^2 = 1$.

$$\begin{aligned} D_u f(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{f(\overline{op} + h\bar{u}) - f(\overline{op})}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h} \end{aligned}$$

Remarks : (i) $f_x = Df_i$ $i = \langle 1, 0 \rangle$

(ii) $f_y = Df_j$ $j = \langle 0, 1 \rangle$

Theorem : How to compute $D_u f(x_0, y_0)$

$$\begin{aligned} D_u f(x_0, y_0) &= \langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle \cdot \langle a, b \rangle \\ &= a f_x(x_0, y_0) + b f_y(x_0, y_0) \end{aligned}$$

Why : ? Let $g(h) = f(x_0 + ah, y_0 + bh)$

Then $D_u f(x_0, y_0) = g'(0)$ Now,

$$\begin{aligned} g'(h) &= f_x a + f_y b \\ \Rightarrow g'(0) &= f_x(x_0, y_0) a + f_y(x_0, y_0) b \end{aligned}$$

(2) • The gradient vector (梯度向量) of f at (x_0, y_0)

定義 $\langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle =: \nabla f$

• $D_u f(x_0, y_0) = \text{gradient 和 } u \text{ 的內積} .$

Example 1 : $f(x, y) = x^3 - 3xy + 6y^2$. Find the directional derivative of f at $(1, 2)$ in the direction of the vector $\langle \sqrt{3}, 1 \rangle$.

Solution :

$$\begin{aligned} D_u f(1, 2) &= \left\langle 3x^2 - 3y, -3x + 12y \right\rangle \Big|_{\substack{x=1 \\ y=2}} \cdot \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle \\ &= \frac{-3\sqrt{3} + 21}{2} \end{aligned}$$

(3) Facts

(i) $D_u f = \nabla f \cdot u = f$ 在 u 方向的變化率

(ii) $\max_u D_u f = |\nabla f|$ when $u // \nabla f$ (same direction)

$$\because \nabla f \cdot u = |\nabla f| |u| \cos \theta, |u| = 1$$

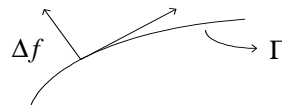
$$= |\nabla f| \quad \text{if } \theta = 0$$

(iii) $\min_u D_u f = -|\nabla f|$. When $u // \nabla f$ (opposite direction).

(iv) (a) $\nabla f \perp$ level curve

(b) $\nabla f \perp$ 任何在 level surface 的曲線.

(a) Consider $\Gamma: f(x(t), y(t)) = C$



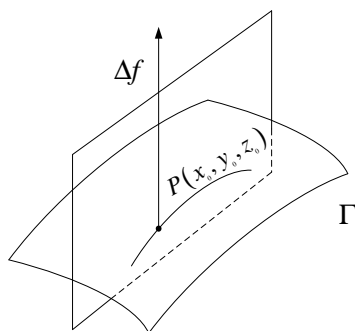
$$\Rightarrow f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = 0$$

$$\Rightarrow \nabla f \cdot \left(\frac{dx}{dt}, \frac{dy}{dt} \right) = 0$$

(b) Consider $\Gamma: f(x, y, z) = C$. Let

$f(x(t), y(t), z(t)) = C$ be any curve on the level surface $\Gamma: f(x, y, z) = C$.

Then $\nabla f \cdot \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) = 0$.



(v) Consider surface $\Gamma: z = f(x, y)$. Let $g(x, y, z) = f(x, y) - z = 0$ 則 Γ 可

當成 g 函數的一個 level surface. 此 g 函數的 gradient 為 $\nabla g = \langle f_x, f_y, -1 \rangle$.

(即將二維函數看成三維函數的等高曲面).

由(iv-b) ∇g 為過點 $P(x_0, y_0, z_0)$ 和 Γ 相切的平面的法向量.

Example 3 : Given a level surface : $\frac{x^2}{4} + \frac{y^2}{1} + \frac{z^2}{9} = 3$. 求過 $(-2, 1, -3)$ 且此等高曲面相切的切面方程式.

Solution :

$$\vec{n} = \left\langle \frac{x}{2}, 2y, \frac{2z}{9} \right\rangle \bigg|_{\substack{x=-2 \\ y=1 \\ z=-3}} = \left\langle -1, 2, -\frac{2}{3} \right\rangle$$

$$\text{切面方程式 } \left\langle -1, 2, -\frac{2}{3} \right\rangle \cdot \langle x+2, y-1, z+3 \rangle = 0$$

$$\Rightarrow -(x+2) + 2(y-1) - \frac{2}{3}(z+3) = 0$$

Example 4 :
$$\begin{cases} f: z = x^2 + y^2 \\ g: 4x^2 + y^2 + z^2 = 9 \end{cases}$$

求過 $(-1, 1, 2)$ 和其交集的曲線相切的切線方程式.

Solution :

此切線的方向和 $\nabla f, \nabla g$ 垂直

$\Rightarrow$ 切線的方向為 $\nabla f \times \nabla g$.

$$\nabla f = \langle 2x, 2y, -1 \rangle = \langle -2, 2, -1 \rangle$$

$$\nabla g = \langle 8x, 2y, 2z \rangle = 2 \langle -4, 1, 2 \rangle$$

$$\langle -2, 2, -1 \rangle \times \langle -4, 1, 2 \rangle = \langle 5, 8, 6 \rangle$$

$$\Rightarrow \frac{x+1}{5} = \frac{y-1}{8} = \frac{z-2}{6}$$